

TED (15)6013

DIPLOMA EXAMINATION IN ENGINEERING/MANAGEMENT/COMMERCIAL PRACTICE

STRUCTURAL DESIGN II

Time 3 hours

(Maximum marks ;100)

(Note- use of IS 800-2007,IS 875,IS1905 and steel Tables are permitted)

PART A (Maximum marks 10)

1. Answer all the question in one or two sentences. Each questions carries 2 marks)

1. Define edge distance
2. Write short notes on HSFG bolt
3. What is lug angles
4. What is plastic cross section
5. Define cavity wall

(5x2=10)

PART B(MAXIMUM MARKS 30)

(Answer any five questions. Each question carries 6 marks)

- II)
- 1 .What are the advantages and disadvantages of steel structur
 2. Write the physical and mechanical properties of steel structures
 - 3 What are the modes of failures of tension member
 - 4 Write the codal provisions of the effective length of column for different end conditions
 - 5 Differentiate between the laterally supported and un supported beam
 - 6 Explain plastic carrying capacity of the beam
 - 7 What are the loads on roof truss 5X6=30

PART C(maximum marks 60)

UNIT 1

(Answer one full question from each unit Each full question carries 15 marks)

III. 1 A tie member of a roof truss consist of ISA 90X60X10 mm, the angles are connected on the either side of 12 mm gusset plate and member is subjected to factored pull of 350 KN. Design the welded connection. Assume the welding is to be made in the workshop. 10 marks

2. What are the advantages of welded connection 5 marks

OR

IV .1. A tie member in a roof truss consist of an ISA 100X75X10MM thic by a shop fillet welds on both sides of angles.Design the fillet weld joints with permissible stress and ultimate shear stress are 150 mpa and 410 mpa 9 marks

2' What are the advantages and disadvantages of bolted connections 6 marks

UNIT 2

V.1 .Determine the tensile strength of a tension member ISA 90X60X10 mm connected to a gusset plate of 12 mm thick . T he length of the weld is 350 mm. Given $f_y=250$ mpa, $F_u=330$ mpa 9marks

2. Explain the component parts of plate girder arks 6

OR

VI. 1. Design a steel column using rolled steel I section to carry a factored load of 500 KN. The length of member is 3 m. Column is fixed at both ends. $F_y= 300$ MPA 9 marks

2.Write any four codal provision of lacing system 6 marks

UNIT 3

VII. 1.A hall measuring (15x6) m consists of beam spaced at 3 m c/c. RCC slab of 120 mm is cast over the beam.T he imposed load is 4 KN/M^2 .The beam is supported on 300 mm wall.Design one intermediate beam and check the design for deflection and shear marks 10

2. What are the types of beam section marks 5

OR

VIII. 1. Design the simply supported beam of effective span 2m carryingthe factored concentrated load of 400 KN at mid span. Marks 9

2 . Determine the plastic moment capacity and plastic section modulus of rectangular beams section of size bxd about ZZ axis marks 6

UNIT 4

IX. 1.Determine the design load on roof truss of a factory building for a span of 20 m and pitch of 1/5 .height of truss at eves level is 10 m.spacing of truss is 4.5 m.Factory building is 36 mlong located at Delhi,proviude AC sheet roofing marks 9

2. Explain the components parts of roof truss marks 6

OR

XI. 1. A masonry wall carrying an axial load of 10 kN/m is of 3.5 m effective length . It is not braced by cross wall. The effective height of wall is 2.4 m. Design the masonry wall marks 9

2 . Explain the terms – effective length, effective height and slenderness ratio of masonry wall marks 6